PASS

# Modernize with SQL Server 2019 and Intel® Technologies

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*Presenting Sponsors:*

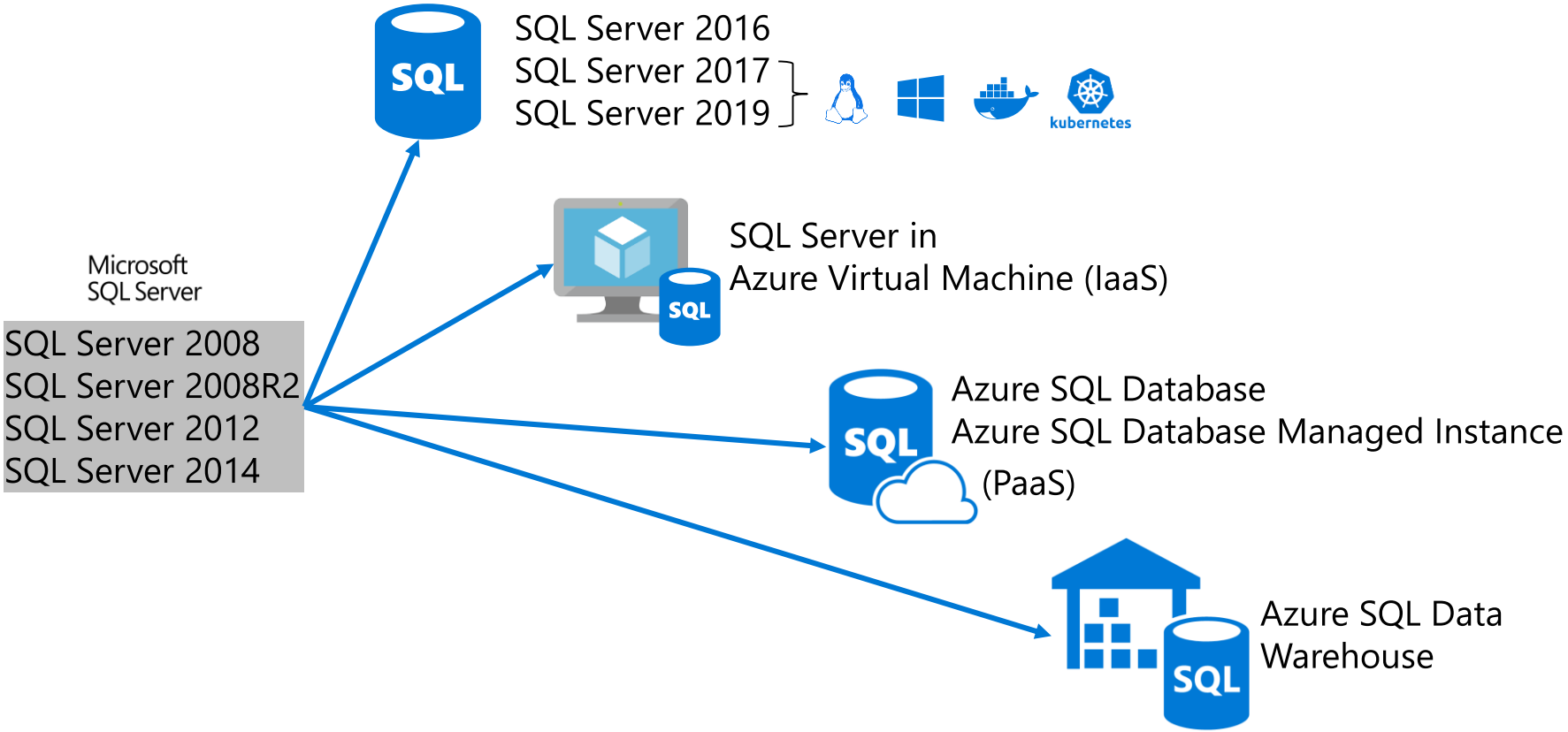
Microsoft intel

# SQL Server 2019:
# The Modern Data Platform

# Modernizing the WideWorldImporters Company

- Mixture of SQL Server 2008, 2014, and 2016 servers
- Expensive ETL applications for data sources outside of SQL Server
- Should we use “Big Data” technologies?
- Query performance tuning expensive
- Some database applications suffer from I/O performance
- Need data secure end-to-end with classification and auditing
- Applications need better availability and less downtime
- Our company uses a mixture of different operating system platforms
- Popularity of containerized applications growing
- What, when, and how should I migrate to the cloud?
- Desire to build more intelligent applications with AI and Machine Learning

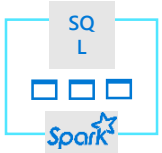
# Azure Data Modernization Choices



# Modernize with SQL Server 2019

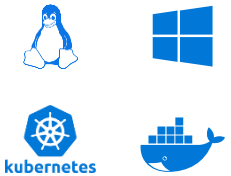
## Now with big data clusters

### Intelligence over any data




Analytics over structured and unstructured data with the power of SQL and Apache Spark

### Choice of platform and language



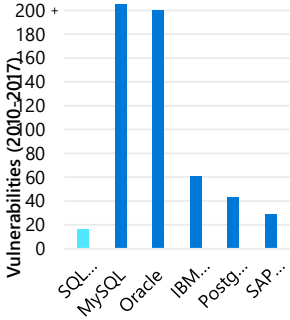

| | | |
|---|---|---|
| T-SQL | SCALA | Python |
| Java | Node.js | Ruby |
| C/C++ | C#/VB.NET | .NET core |

### Industry-leading performance

#1 OLTP performance[1]

#1 DW performance on 1TB[2], 10TB[3], and 30TB[4]

### Most secure over the last 8 years[5]




### Insights in minutes and rich reports

The best of Power BI and SQL Server Reporting Services in Power BI Report Server

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Private cloud

In-memory across all workloads

Most consistent data platform

1/10th the cost of Oracle

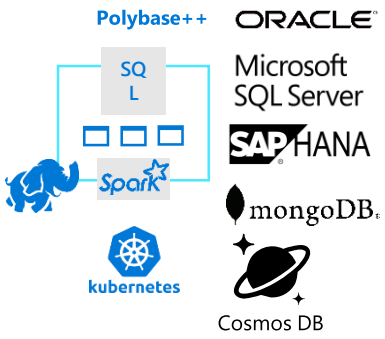
Public cloud

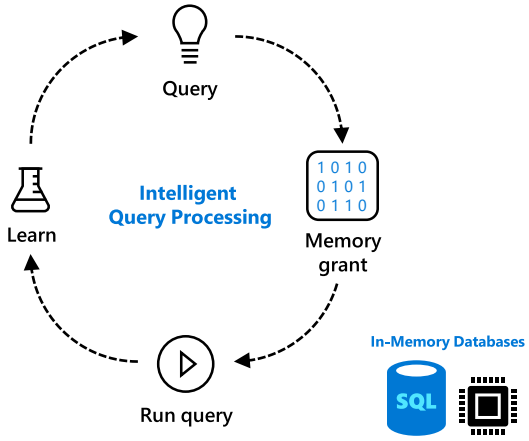
All TPC Claims as of 1/19/2018.

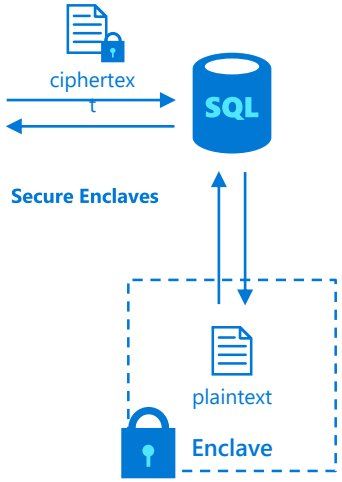
[1] http://www.tpc.org/4081; [2] http://www.tpc.org/3331; [3] http://www.tpc.org/3326; [4] http://www.tpc.org/3321; [5] National Institute of Standards and Technology Comprehensive Vulnerability Database

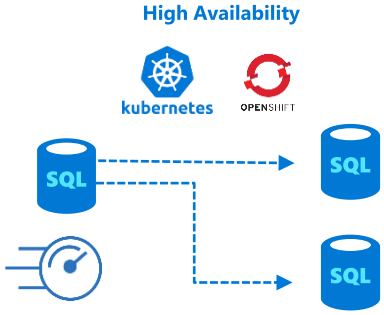
# SQL Server 2019

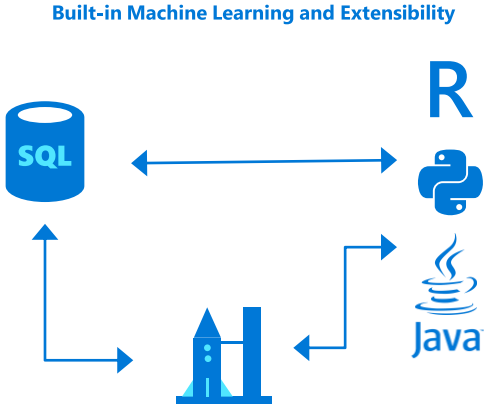
## Key New Functionality

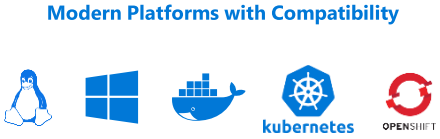


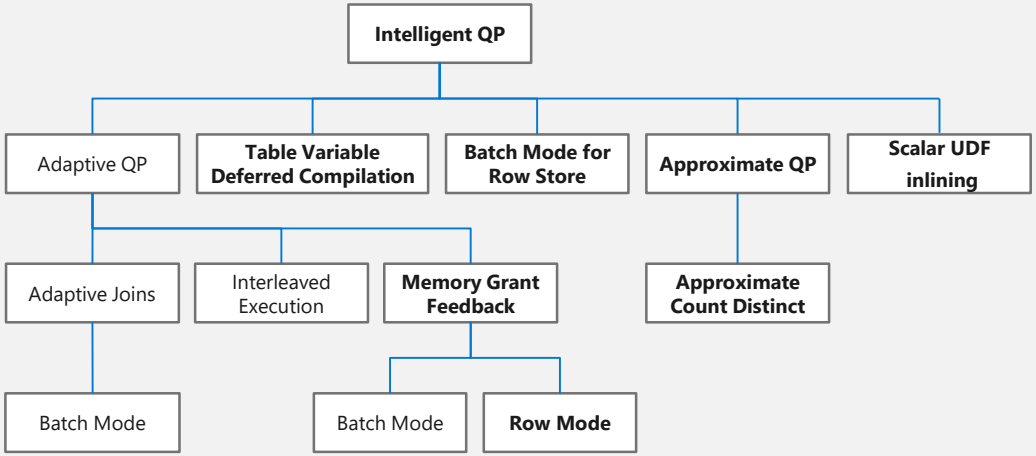



# Intelligent Performance with SQL Server

## The intelligent database

- **Intelligent Query Processing**
- Performance insights anytime and anywhere with **Lightweight Query Profiling**
- Accelerating I/O performance with **Persistent Memory**
- TempDB: It Just Runs Faster **Memory-optimized TempDB Metadata**

### The Intelligent Query Processing feature family




Bold indicates new and improved features in SQL Server 2019

# Mission critical security

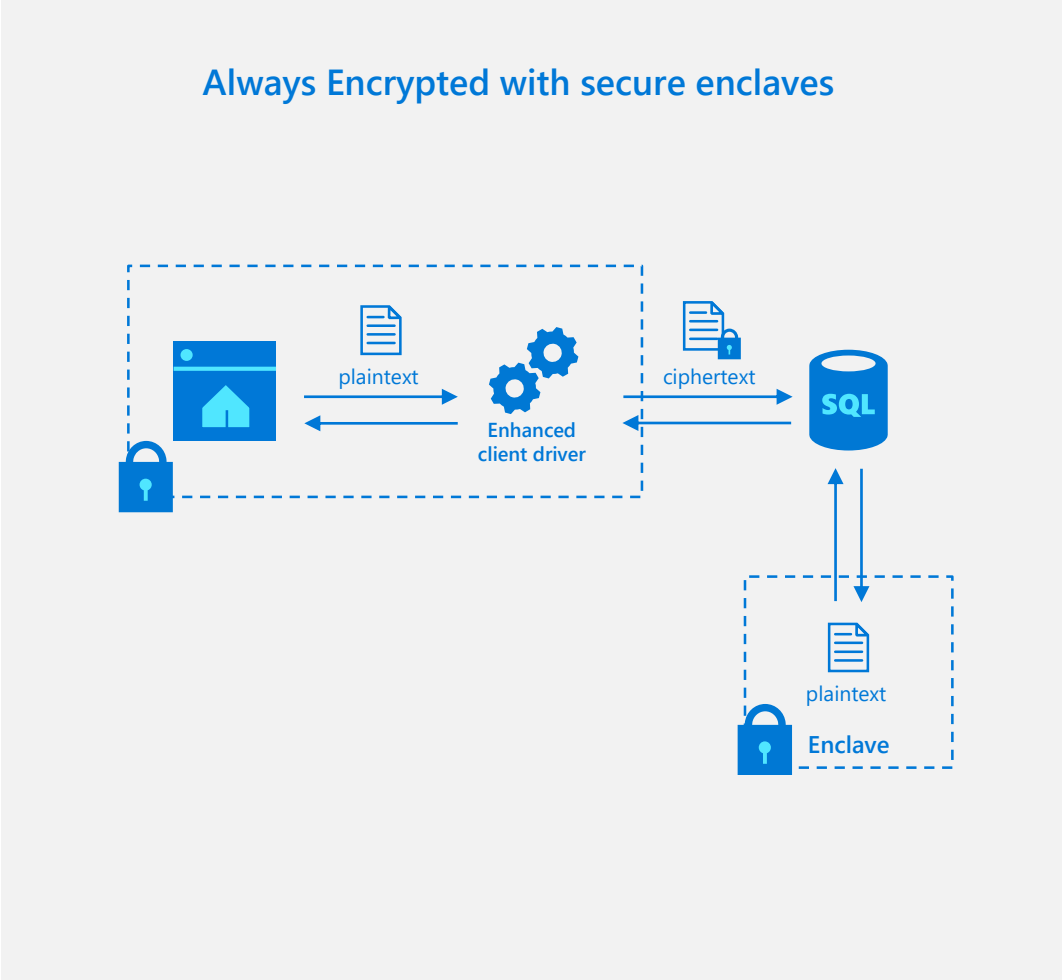
## Confidential computing

**Always Encrypted** with secure enclaves

**Data Classification** and auditing built-in

TDE scan suspend and resume

Simplified certificate management



# High Availability

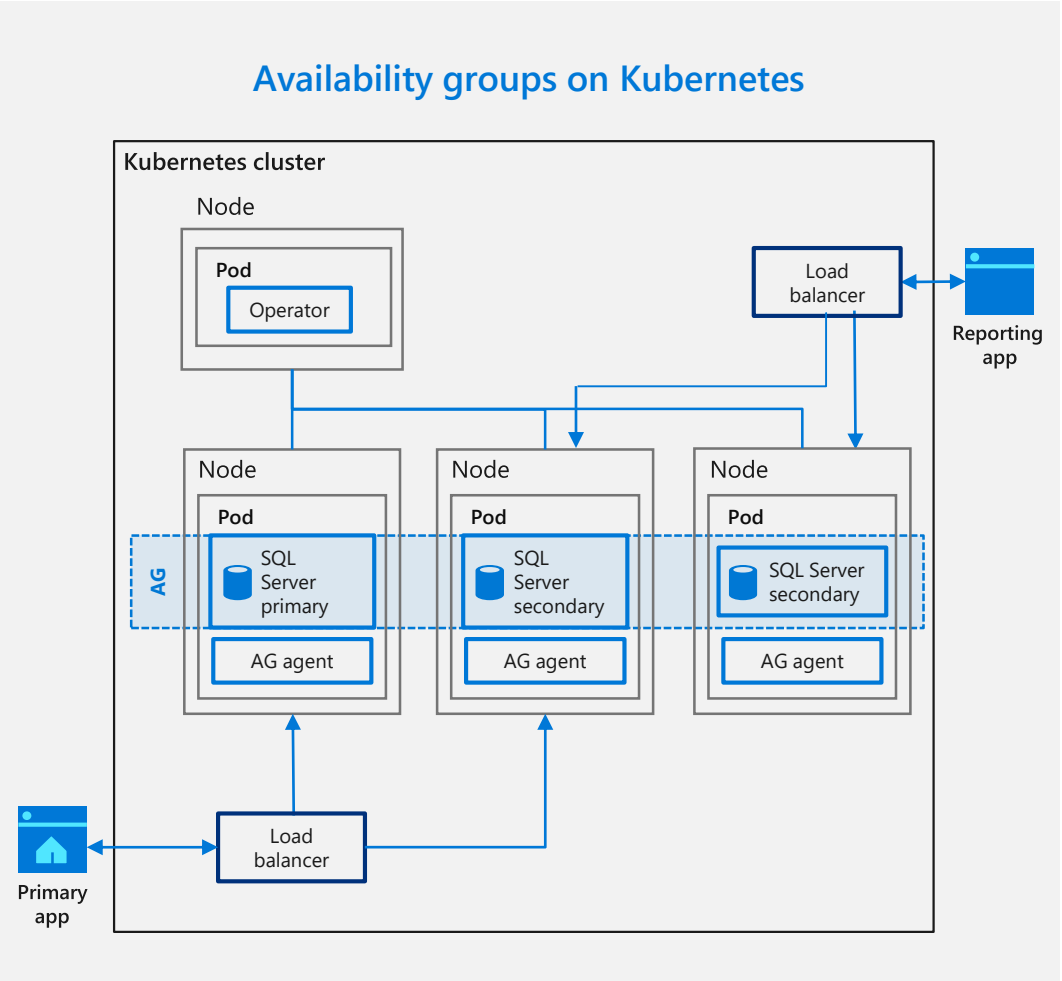
## Keep SQL Server running

**Online Index** Enhancements

Availability Groups Enhancements
**System Databases (Planned)**

Availability groups on **Kubernetes**

**Accelerated Database Recovery**



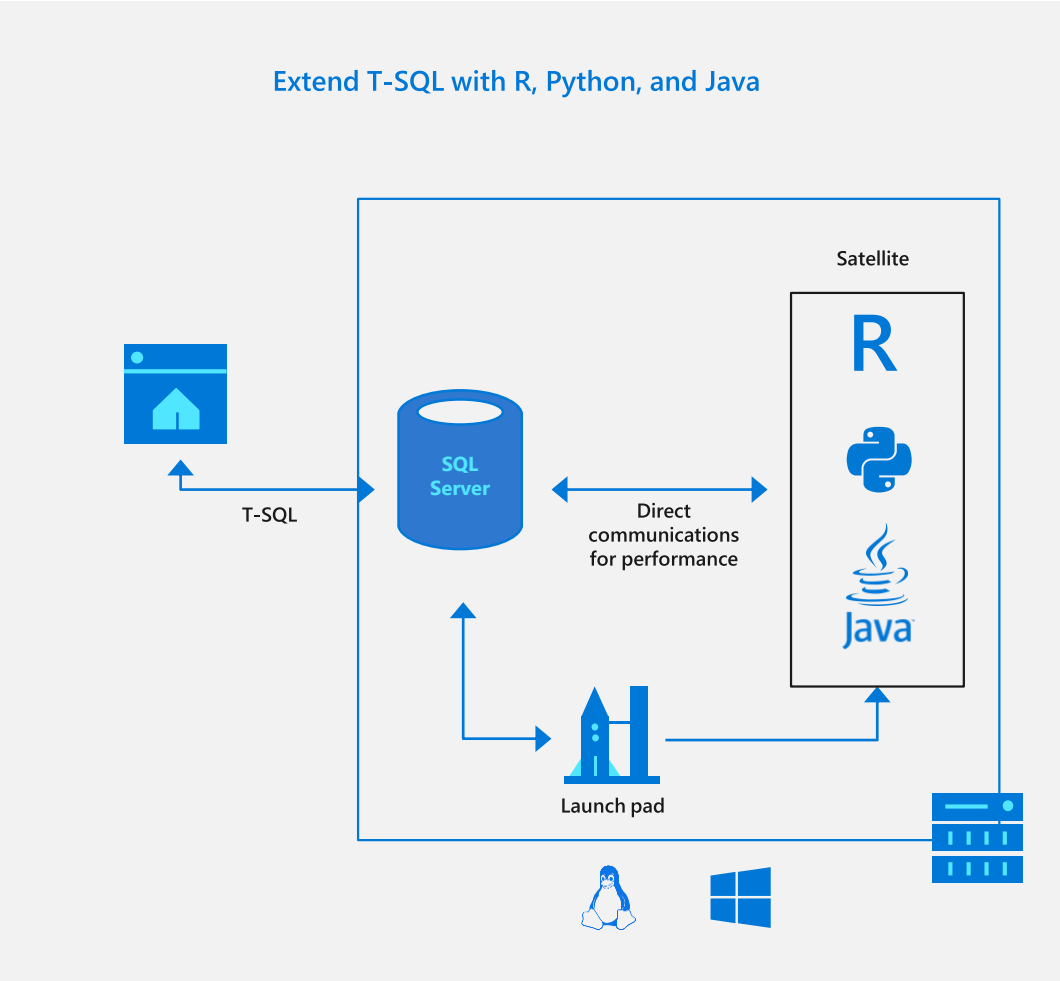
# Enhancing the developer experience

**SQL Graph** enhancements

**UTF-8** support

**Machine Learning Services** enhancements

SQL Server **Java** extension



# SQL Server 2019 and Linux and Containers

**New Features**

- Replication
- Distributed transactions
- Machine Learning
- Polybase

The **Microsoft Container Registry**

Red Hat Images

Availability Groups on **Kubernetes**

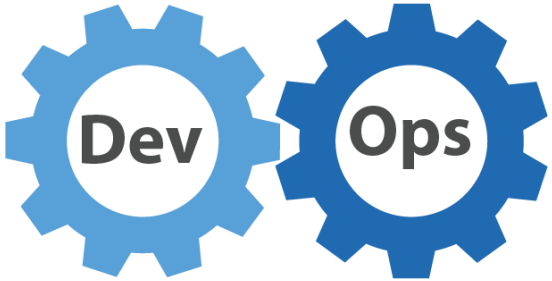
# Why SQL Server and Containers?

**Portable**
Run anywhere Docker is supported

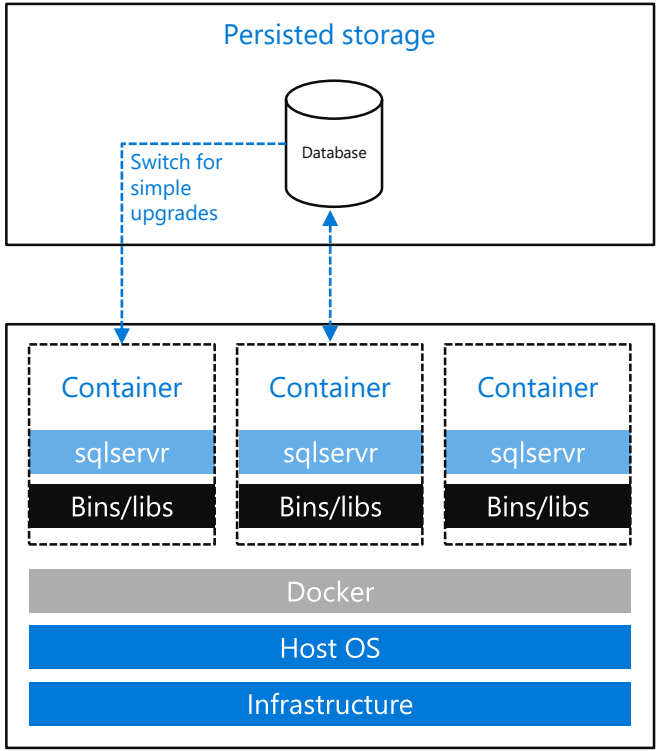
**Lightweight**
Reduced disk, CPU, and memory footprint

**Consistent**
Consistent image of SQL Server, scripts, and tools

**Efficient**
Faster deployment, reduced patching, and less downtime

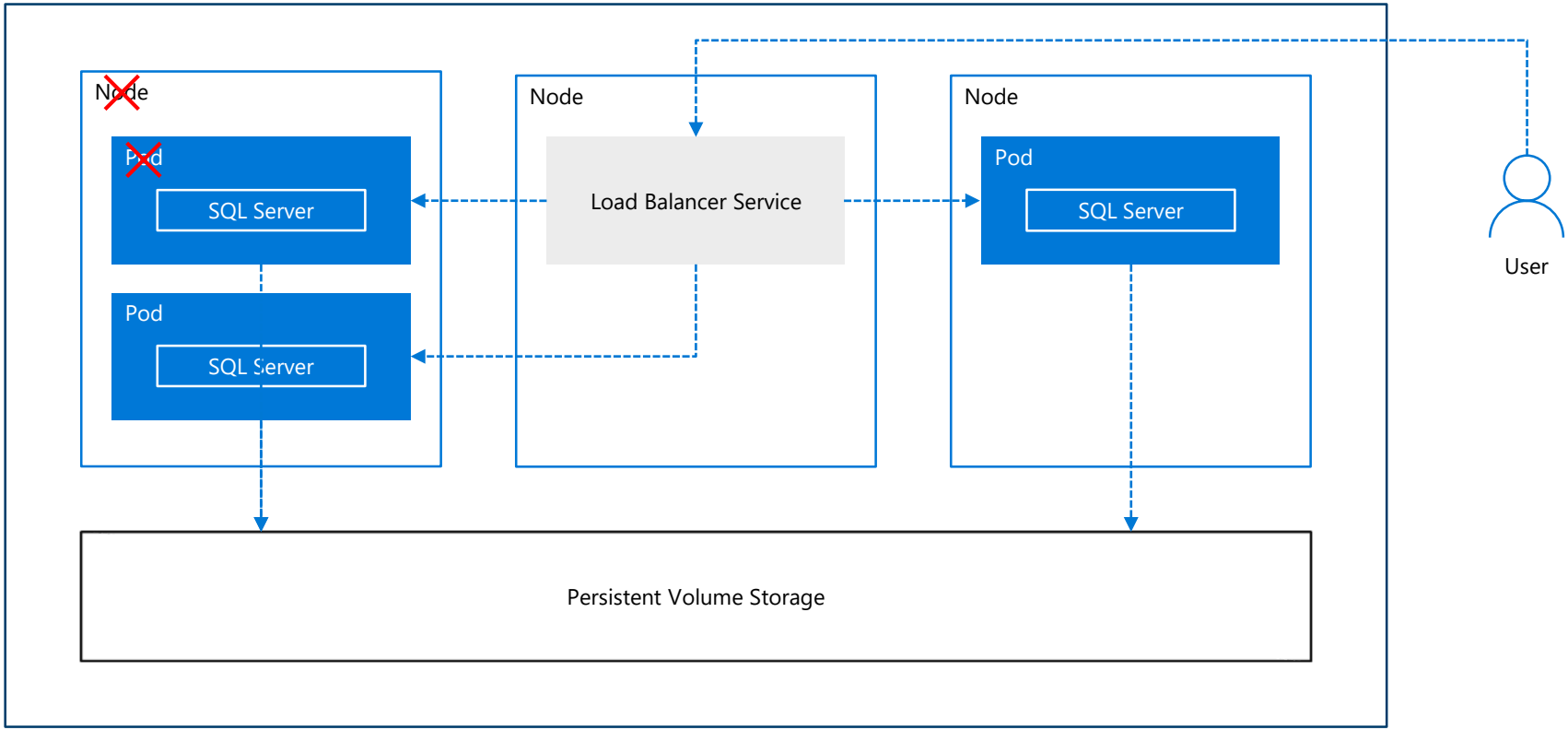



**Container configuration**



# SQL Server - Shared storage HA in Kubernetes

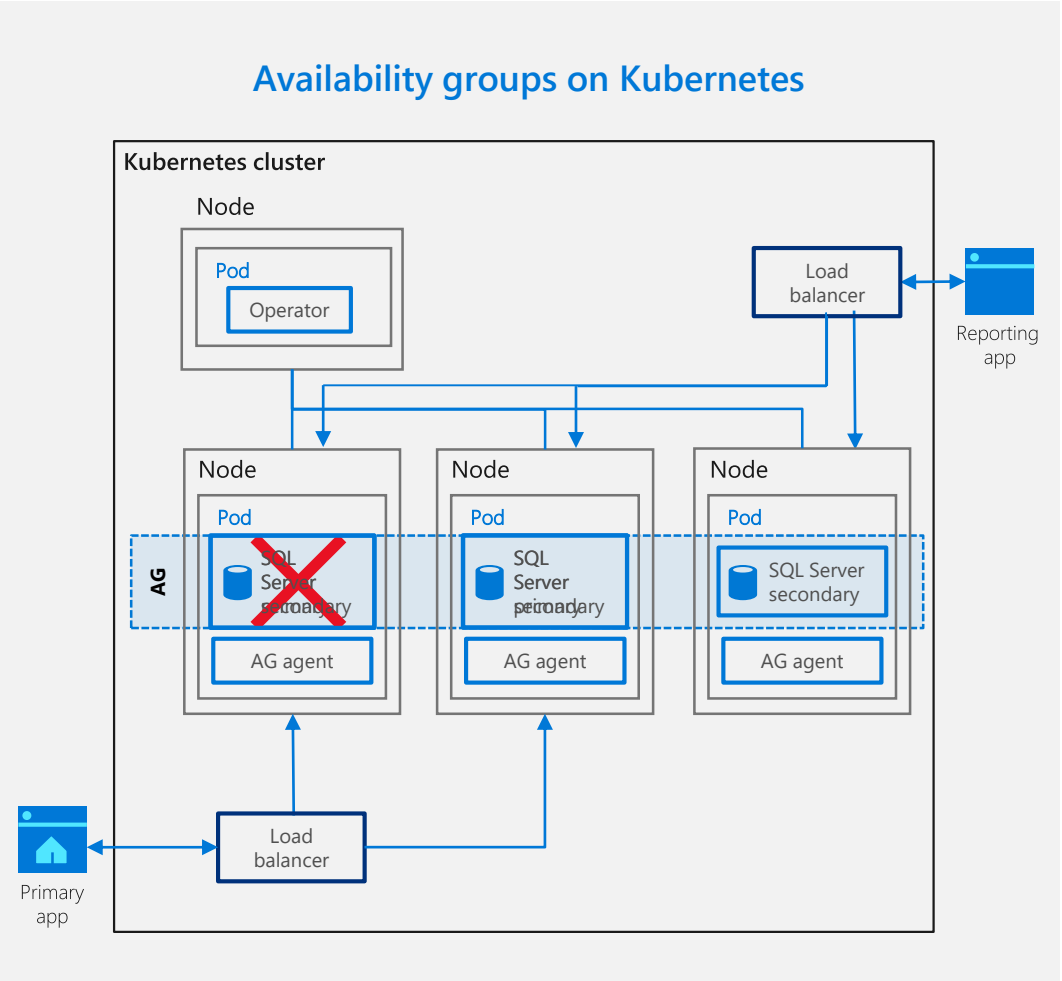
## Built-in HADR orchestration with no clustering required



# SQL Server 2019

# Always On Availability Groups on Kubernetes

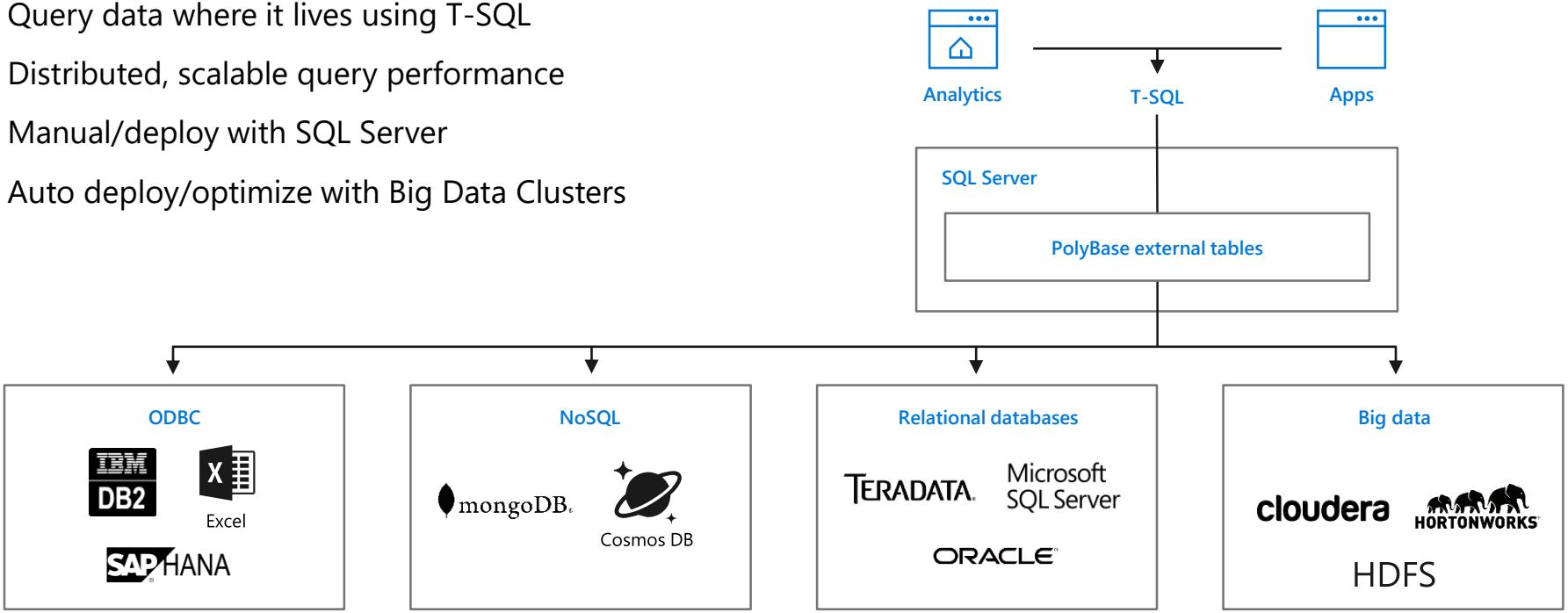
- SQL Server/k8s failover integration
- Operator deployment
- AG concepts all apply
- Load Balancer for Primary App
- Load Balancer for Secondary Replica Readers



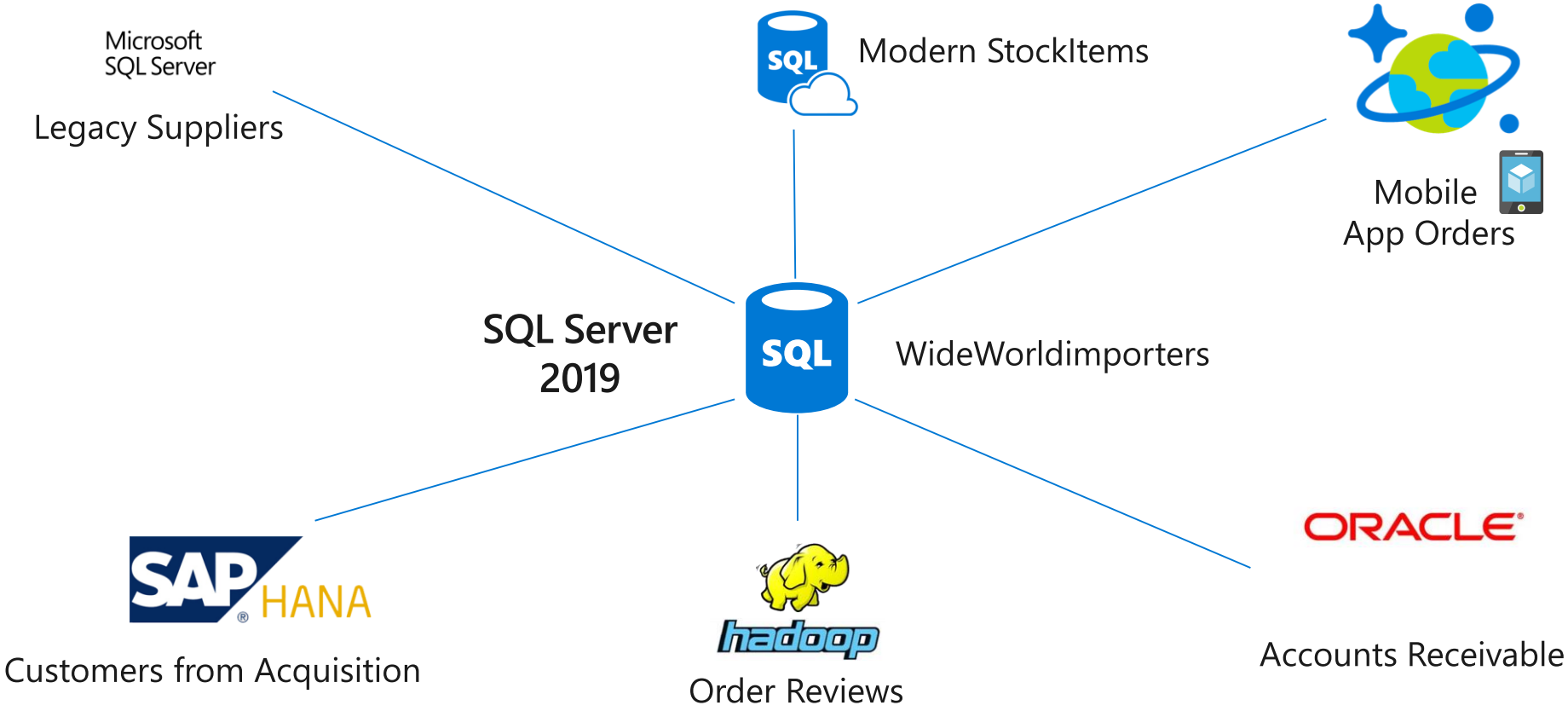
# What is SQL Server Polybase?

"It's all about Data Virtualization"

- ✓ Distributed compute engine integrated with SQL Server
- ✓ Query data where it lives using T-SQL
- ✓ Distributed, scalable query performance
- ✓ Manual/deploy with SQL Server
- ✓ Auto deploy/optimize with Big Data Clusters

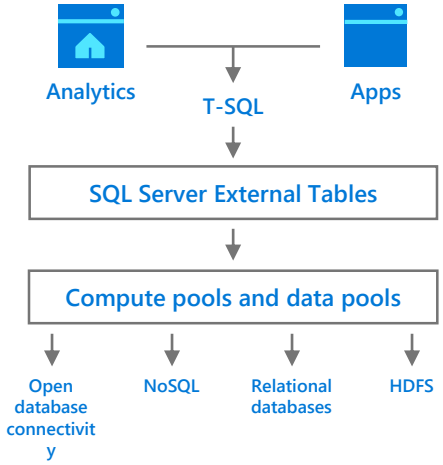


# SQL Server 2019: Data Virtualization



# SQL Server 2019 Big Data Clusters

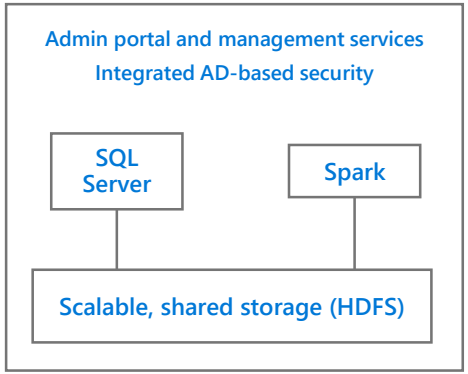
## Data virtualization




Combine data from many sources without moving or replicating it

Scale out compute and caching to boost performance

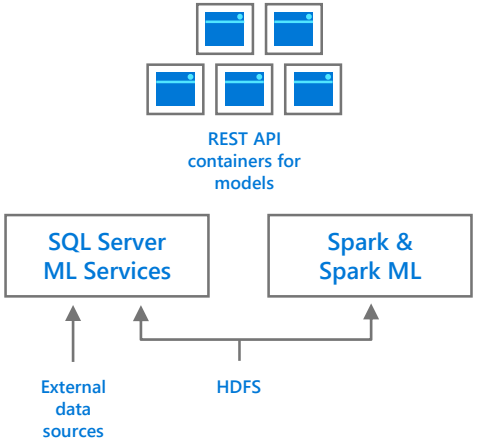
## Managed SQL Server, Spark, and data lake




Store high volume data in a data lake and access it easily using either SQL or Spark

Management services, admin portal, and integrated security make it all easy to manage

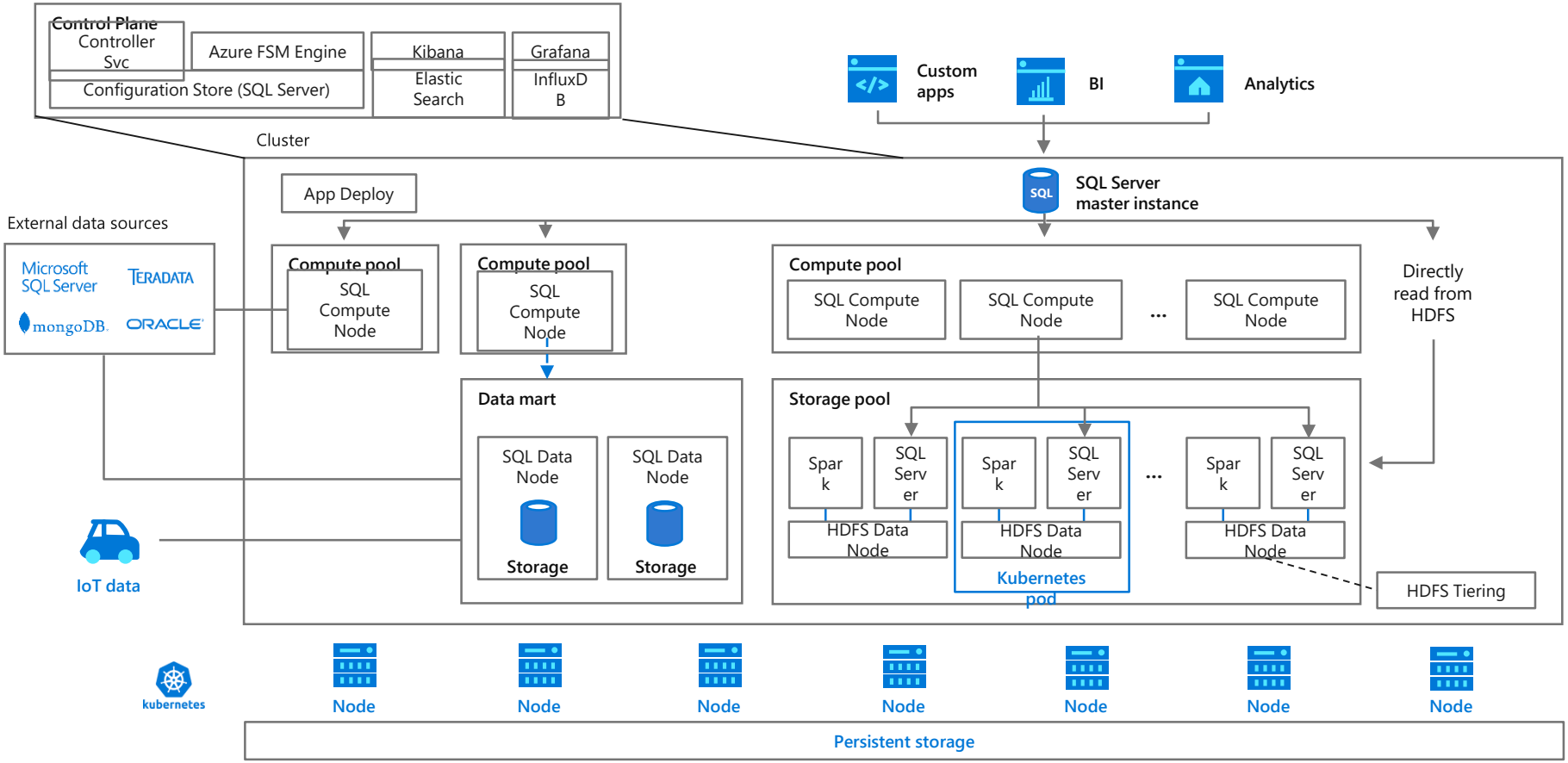
## Complete AI platform




Easily feed integrated data from many sources to your model training

Ingest and prep data and then train, store, and operationalize your models all in one system

# SQL Server Big Data Cluster Architecture



# The Customer Voice

- Columnstore stats in DBCC CLONEDATABASE
- Estimate compression for Columnstore indexes
- Diagnostics for auto stats blocking
- The #1 voted customer feedback item of all time: String Truncation (1000+ votes)
- Troubleshoot page resource waits with new built-in T-SQL

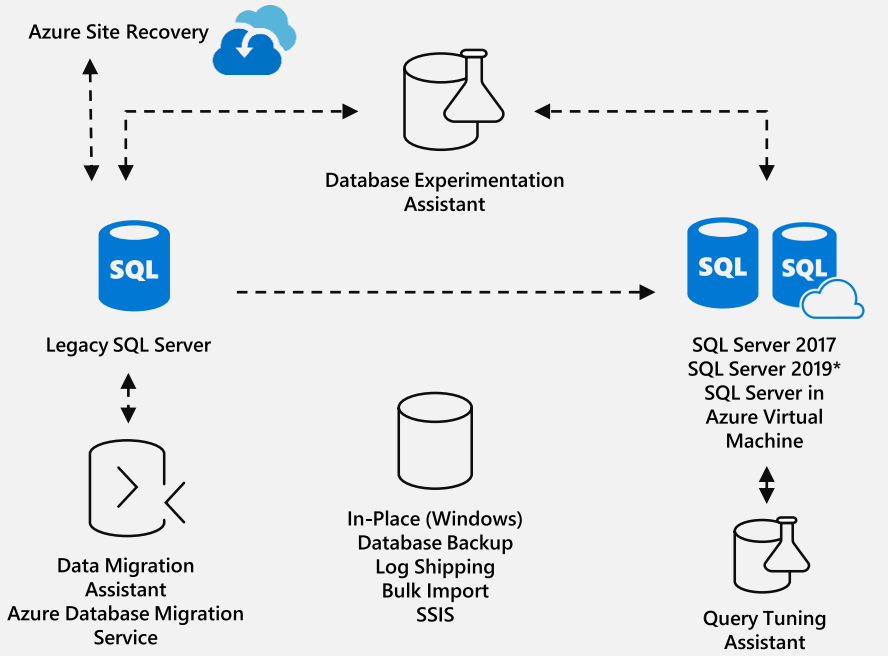
```
String or binary data would be truncated
```

```
String or binary data would be truncated
in table '%.*ls', column '%.*ls’.
Truncated value: '%.*ls'
```

```
SELECT page_info.*
FROM sys.dm_exec_requests AS d
  CROSS APPLY
sys.fn_PageResCracker(d.page_resource) AS r
  CROSS APPLY sys.dm_db_page_info(r.db_id,
r.file_id, r.page_id,'DETAILED')
    AS page_info;
```

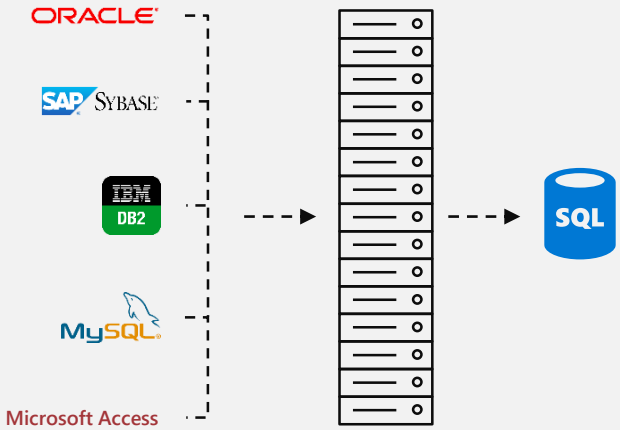
# Migrate to the Modern SQL Server

## Migration from legacy SQL Server




* Coming by GA

## Migration from external databases




SQL Server Migration Assistant (SSMA)

# Learn more

http://aka.ms/bobwardms
http://aka.ms/bobsqldemos
http://aka.ms/sqllinuxbook

Use our free training at https://aka.ms/sqlworkshops

Learn from videos and demos at https://aka.ms/sqlchannel

Download and try it at http://aka.ms/ss19

What's new for SQL 2019 documentation

Sign-up for the EAP program at https://aka.ms/eapsignup

# SQL Server 2008 EOS Is Coming - Modernizing with Intel® Technologies

# Data-Centric Infrastructure Focus

## MOVE Faster

- intel ETHERNET
- intel SILICON PHOTONICS
- intel OMNI-PATH FABRIC

## MOVE More

- intel OPTANE™ DC SOLID STATE DRIVE
- intel OPTANE™ DC PERSISTENT MEMORY

## PROCESS Everything

Software & System-Level **OPTIMIZED**

intel select solution

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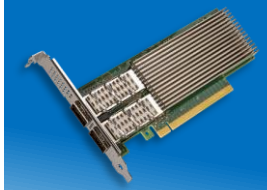
# Data-Centric Portfolio

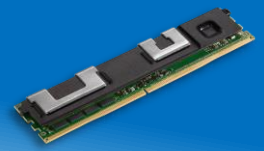
## MOVE Faster

INTEL® ETHERNET 800 SERIES



## STORE More

INTEL® OPTANE™ DC Persistent Memory



DUAL Port INTEL® OPTANE™ DC SSD

INTEL® QLC 3D NAND SSD

## PROCESS Everything

2nd Generation INTEL® Xeon® Scalable

INTEL® Xeon® D-1600

INTEL® AGILEX™ FPGA

intel select solution

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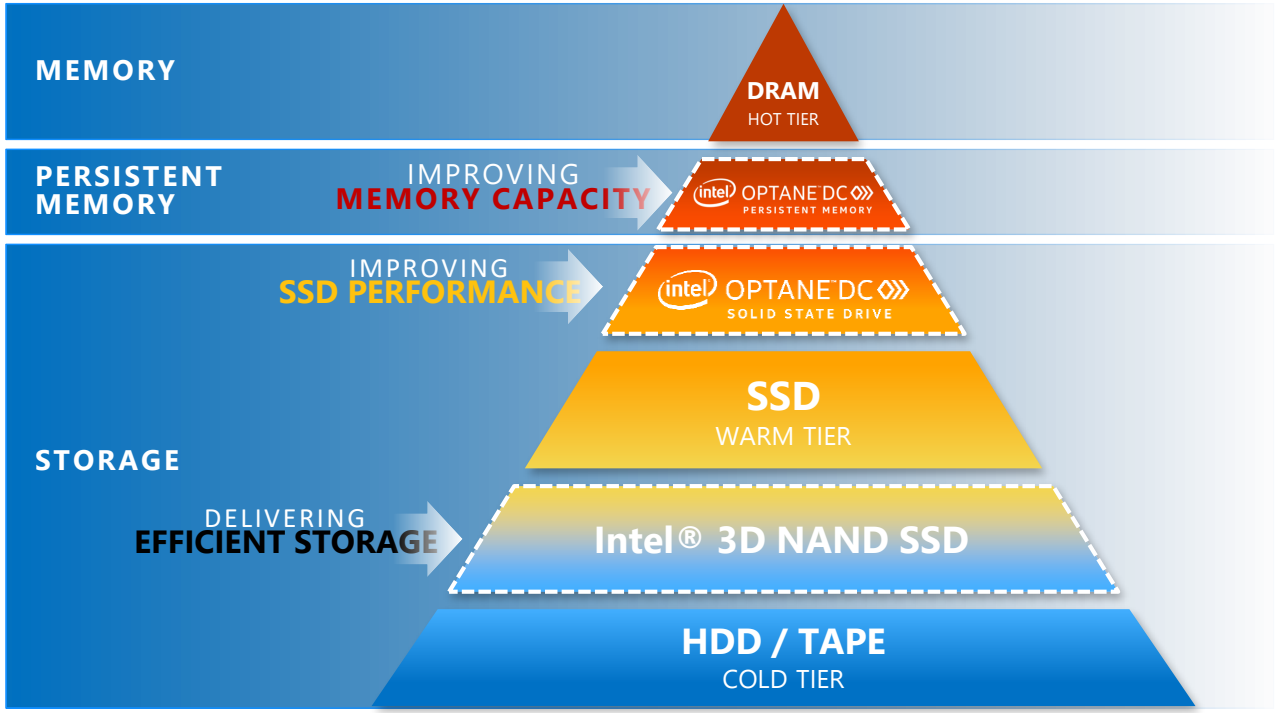
# Transforming the Memory Storage Hierarchy

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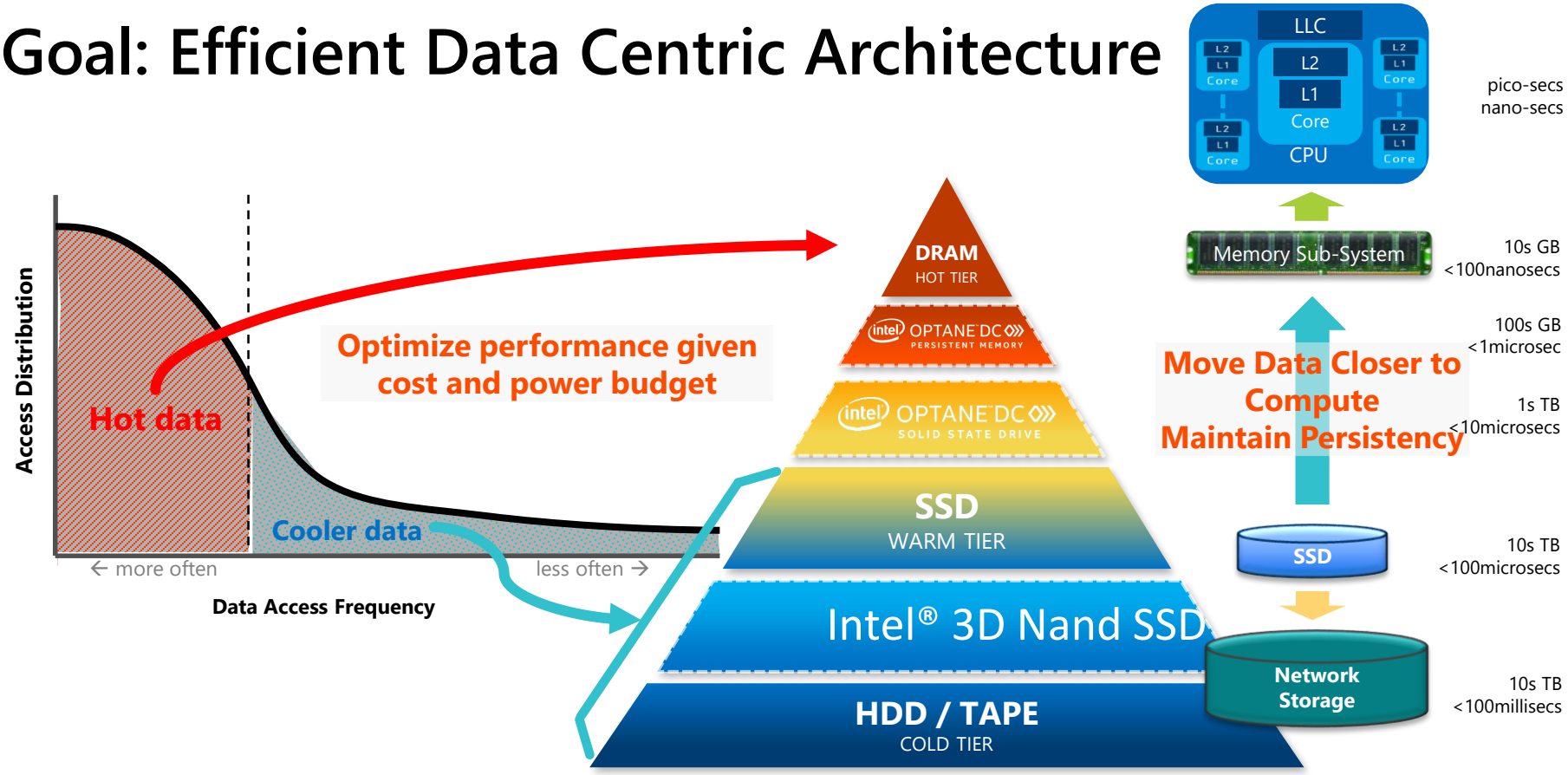
# Goal: Efficient Data Centric Architecture

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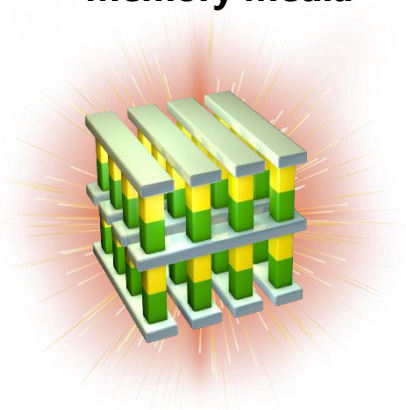
# Technology Driven: Intel® Optane™ Technology

Unleashing breakthrough performance for a new generation of computing

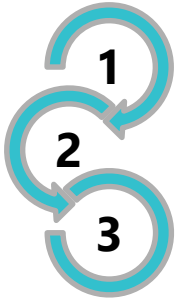
**Intel® 3D XPoint™ Memory Media**



Breakthrough material advances for scalability & high performance

+

**Intel® building blocks**

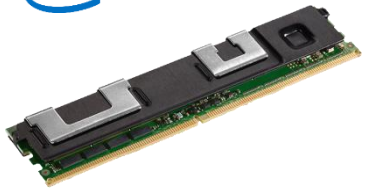



Intel System Memory Controller, Interface HW, and SW IP

=

**Intel® Optane™ Technology**

intel OPTANE™



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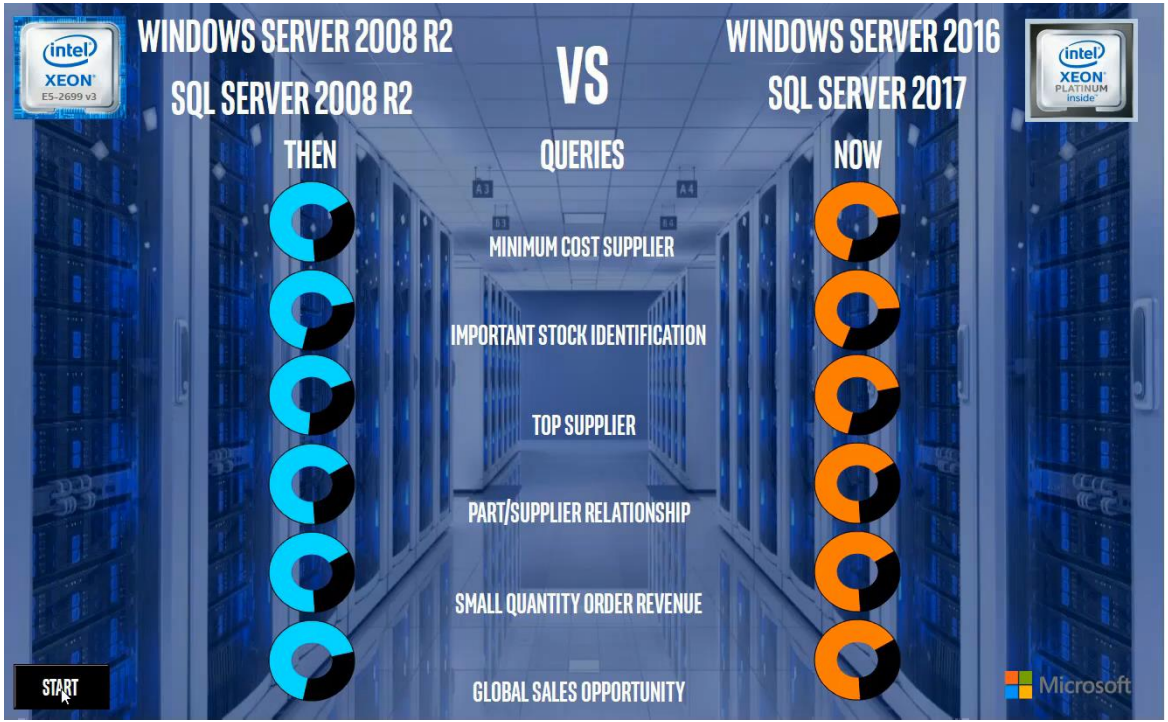
# Reduced Query Runtimes[4]

## With 2nd Gen Intel® Xeon® Scalable Processor + SQL Server 2017




Performance results are based on testing by Intel as of 3/13/2019 and may not reflect all publicly available security updates. See configuration disclosures for details. No product or component can be absolutely secure. Software and workloads used in performance tests may have been optimized for performance only on Intel microprocessors. Performance tests, such as SYSmark and MobileMark, are measured using specific computer systems, components, software, operations and functions. Any change to any of those factors may cause the results to vary. You should consult other information and performance tests to assist you in fully evaluating your contemplated purchases, including the performance of that product when combined with other products. For more complete information visit www.intel.com/benchmarks

[4] Configuration: See Notices & Disclaimers

# Faster Query Performance with 2nd Gen Intel® Xeon® Processor + SQL Server

FASTER DATA WAREHOUSE QUERY PERFORMANCE WITH LATEST SW & HW

PAST CUSTOMER EXPERIENCE

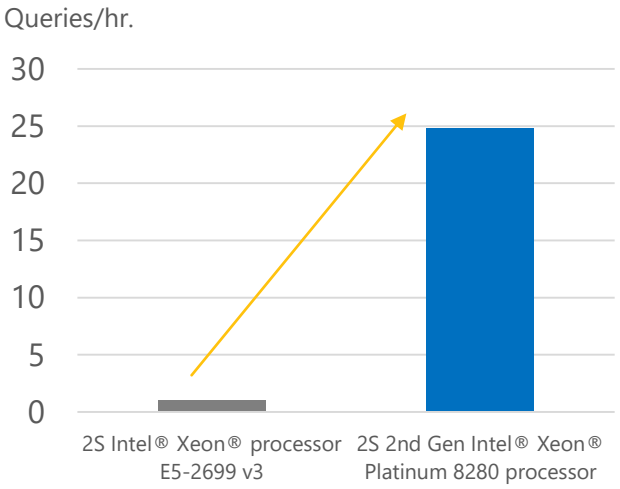
33,681 queries per hour
at 1TB scale factor with a 4 year old system

CUSTOMER EXPERIENCE TODAY

**With 2nd Gen Intel® Xeon® Scalable processors:**

836,261 queries per hour
at 1TB scale factor

24.8X better

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*Other names and brands may be claimed as the property of others

# Memory Mode Example- Virtualized SQL Server

INCREASE VMS PER NODE FOR MULTI-TENANT VIRTUALIZED DATABASES

ALL DRAM CONFIGURATION (768 GB)

22 Microsoft SQL VM instances with DRAM memory at ~$1,588 USD per VM

DRAM (192 GB) + INTEL® OPTANE™ DC PERSISTENT MEMORY (1024 GB)

30 Microsoft SQL VM instances at ~$1,108 USD per VM

36% more VMs per node[1]

30% lower estimated HW cost per VM[2]

Performance results are based on testing by Intel as of 3/13/2019 and may not reflect all publicly available security updates. See configuration disclosures for details. No product or component can be absolutely secure. Software and workloads used in performance tests may have been optimized for performance only on Intel microprocessors. Performance tests, such as SYSmark and MobileMark, are measured using specific computer systems, components, software, operations and functions. Any change to any of those factors may cause the results to vary. You should consult other information and performance tests to assist you in fully evaluating your contemplated purchases, including the performance of that product when combined with other products. For more complete information visit www.intel.com/benchmarks

[1-2] Configuration: See Notices & Disclaimers

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# Intel® Select Solutions

**BOOST IT Velocity**

WITH PRE-DEFINED AND VERIFIED INFRASTRUCTURE SOLUTION STACKS AVAILABLE FROM A RANGE OF PARTNERS

**>50** VERIFIED PARTNER SOLUTIONS

**INCREASE System Level Performance**

WITH HIGHLY OPTIMIZED CONFIGURATIONS OF INDUSTRY-LEADING INTEL DATA CENTER TECHNOLOGIES

intel select solution

http://www.intel.com/selectsolutions

**UNPARALLELED HW / SW Ecosystem**

TO ACCELERATE YOUR PACE OF INNOVATION ACROSS CRITICAL BUSINESS WORKLOADS

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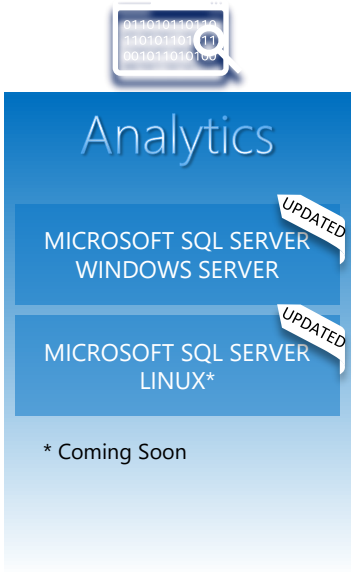
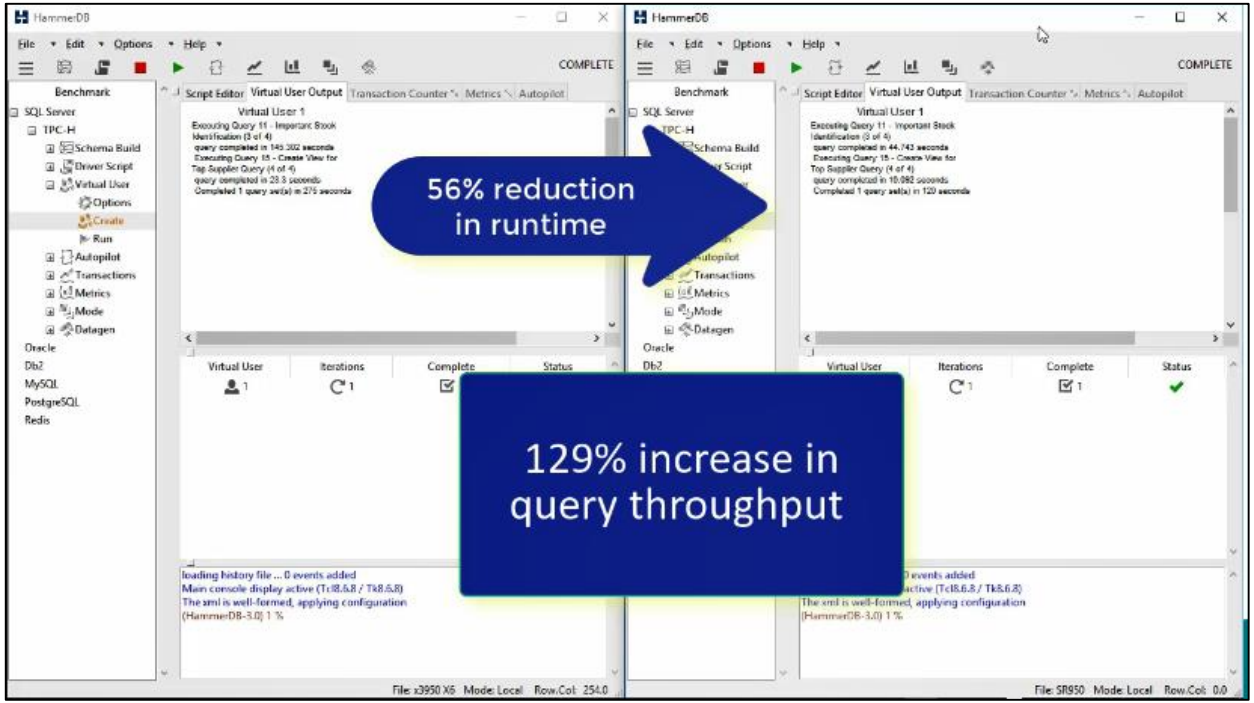
# Making workload acceleration Easy With Intel® Select Solutions



# DAX Example- 30 TB Data Warehouse[5]

## With 2nd Gen Intel® Xeon® Scalable Processor, Intel® Optane™ DC Persistent Memory + SQL Server 2019 CTP

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[5] Configuration: See Notices & Disclaimers

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# Microsoft Ignite ’18 Demo

## The very latest from Intel

**12** x Intel® S2600WFT

✓ **384 GB** (12 x 32 GB) memory

✓ **2 x 28-core** 8260 Intel® Xeon® Scalable processor

✓ **1.5 TB** Intel® Optane™ DC Persistent Memory (as cache)

✓ **32 TB** NVMe (4 x 8TB Intel® DC P4510) as capacity

✓ **2** x 25 Gbps RDMA NICs

Microsoft Ignite
September 24–28, 2018 | Orlando, FL

**Maximize IOPS** read/write 4 kB random, 90% read, 10% write 9,459,587 IOPS
**Maximize throughput** 2 MB sequential, 100% read 535.86 GB/s

## The very latest from Microsoft

https://blogs.technet.microsoft.com/filecab/2018/10/30/Windows-server-2019-and-intel-optane-dc-persistent-memory/

✓ **Windows Server 2019**

✓ Insider Preview, build 17713

✓ Hyper-V + Storage Spaces Direct

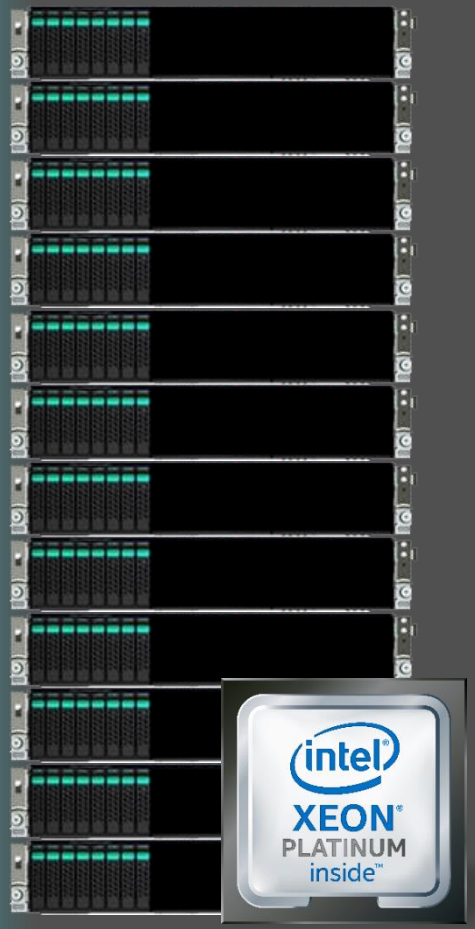
NEW IOPS RECORD

1 3 , 7 9 8 , 6 7 4

Monday, September 24, 2018 | Windows Server 2019 with Intel® Optane™ DC persistent memory

intel Microsoft

**Maximize IOPS, all-read 4 kB random, 100% read**

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1, 2 36% more VMs per node & 30% lower estimated cost per VM. Tested by Intel Corp as of 1/31/2019. 1-node, 2x Intel® Xeon® Platinum 8272L Processor with 768 GB (24 slots / 32 GB / 2666) total memory, Windows Server 2019, 1x Samsung PM963 M.2 960GB, 7x Samsung PM963 M.2 960GB, 4x Intel SSD DC S4600 (1.92TB), OLTP Cloud Benchmark, result: 22 VMs per node, vs. 1-node, 2x Intel® Xeon® Platinum 8272L Processor with 1216 GB (12 slots / 16 GB / 2666 DDR4 memory + 8 slots / 128 GB Intel® Optane™ DC persistent memory) total memory, Windows Server 2019, 1x Samsung PM963 M.2 960GB, 7x Samsung PM963 M.2 960GB, 4x Intel SSD DC S4600 (1.92TB), OLTP Cloud Benchmark, result: 30 VMs per node. Pricing Guidance as of March 1, 2019. Intel does not guarantee any costs or cost reduction. You should consult other information and performance tests to assist you in your purchase decision.

3 Reduced query runtimes with 2nd Gen Intel® Xeon® Scalable Processor, Intel® Optane™ DC Persistent Memory + SQL Server 2019 CTP. Tested by Lenovo Corp and Intel Corp as of 3/31/2019. Lenovo x3590 X6, 8x Intel® Xeon® E7-8890 v3 processors, 12 TB (192 slots / 64GB / 1600) total memory, on Windows Server 2016 Standard Edition, 16x 6.4TB PCIe SSDs, SQL Server 2016 Enterprise Edition, vs. Lenovo SR950, 8x Intel® Xeon® Platinum 8280L processors, 8 TB (64 slots / 128 GB / 2666) total memory, 16 TB (32 slots / 512 GB) Intel® Optane™ DC persistent memory, on Windows Server 2019 Data Center Edition, SQL Server 2019 CTP 2.3. Workload details: 30 TB Data Warehouse, 1 user issuing a set of 22 DSS queries, 4 queries demonstrative of possible gains visualized.

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# Thank You